Binary Search Recurrence

Topic: TimeComplexity File: p4.pdf

PROBLEM

Search for x in a sorted array by repeatedly discarding half of the remaining range.

ALGORITHM

Compare x with the middle value. Continue in the left half if x is smaller, or in the right half if larger.

RECURRENCE

T(n) = T(n / 2) + O(1), because each step makes one comparison and halves the search range.

EXAMPLE

In [2, 5, 8, 12, 16, 23, 38], searching for 16 checks the middle, then narrows to the right side.

COMPLEXITY

Time: O(log n). Iterative extra space: O(1). Recursive implementation uses O(log n) call-stack space.

REQUIREMENT

Binary search requires the data to be sorted and relies on random access for the usual array complexity.